# FACE MASK PRESENCE CLASSIFICATION

Deep Learning Based Image Classification using MobileNetV2

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B.Tech Deep Learning • Semester V, Section A
Group ID: BT_DL_06 | Project ID: DL-006

# PROJECT TEAM & MEMBER DETAILS

**Group: BT_DL_06 | Project ID: DL-006 | Mentor: Yashraj Sir | Team Leader: SANKALP**

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# Project Overview

WHAT THE SYSTEM DOES

## INPUT

A user selects an image containing a person.

## PROCESS

The trained CNN model analyzes the image and predicts a class.

## OUTPUT

The system displays “With Mask” or “Without Mask” with confidence.

This is a binary image-classification project.
The model is trained using a labeled dataset stored in two folders: mask and no_mask.
The trained model can predict on a new image that was not part of the training set.

# Problem Statement & Objectives

WHY THIS PROJECT?

## PROBLEM STATEMENT

Manually checking every image to identify whether a person is wearing a face mask can be time-consuming. The project explores how deep learning can automate this classification from an image.

## OBJECTIVES

- Build a binary image classifier.
- Use transfer learning to reduce training effort.
- Train and validate the model on labeled images.
- Predict a class and confidence score for a new image.

# Technologies & Tools Used

DEVELOPMENT STACK

## Python

Main programming language

## TensorFlow / Keras

Model building, training and loading

## MobileNetV2

Pre-trained CNN feature extractor

## OpenCV

Image reading, color conversion and resizing

## NumPy

Array operations and input preparation

## Matplotlib

Training accuracy and loss graphs

## VS Code

Code editing and project development

## ImageDataGenerator

Rescaling, augmentation and data split

# Dataset & Pre-processing

HOW IMAGES ARE PREPARED

```
data/
  ├── mask/        # images with face masks
  └── no_mask/     # images without face masks
```

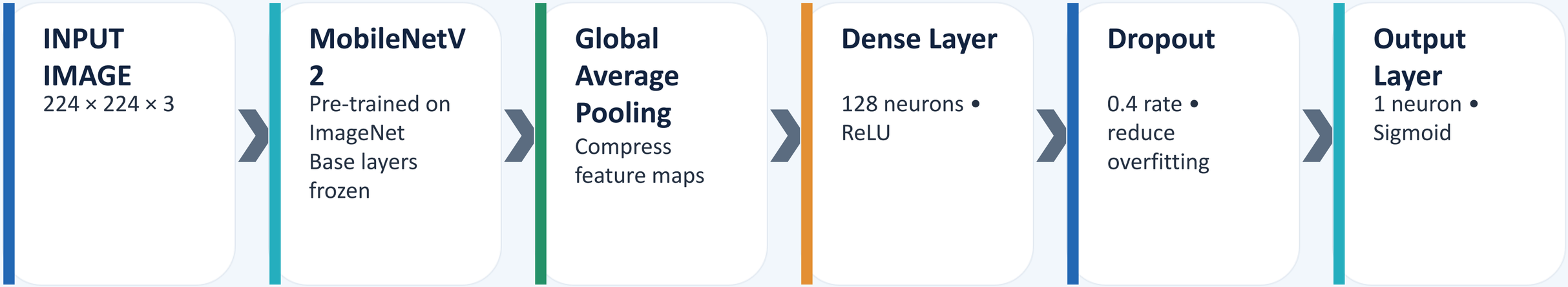
## DATA SPLIT

80% of images are used for training and 20% for validation using validation_split=0.2.

All images are resized to 224 × 224 pixels.
Images are converted to RGB and pixel values are rescaled from 0–255 to 0–1.
Training augmentation: rotation, width/height shift, shear, zoom and horizontal flip.
Augmentation helps the model generalize to small variations in image position and orientation.

# Model Architecture — MobileNetV2

TRANSFER LEARNING




MobileNetV2 provides reusable visual features learned from a large image dataset.
The base model is frozen initially; only the new classification head is trained.
Sigmoid returns a value between 0 and 1 for the binary prediction.

# Model Training

IMPORTANT TRAINING SETTINGS

## EPOCHS

20 maximum training epochs

## BATCH SIZE

32 images per training batch

## OPTIMIZER

Adam • learning rate 0.0001

Loss function: Binary Cross-Entropy, suitable for two-class classification.
Metric: Accuracy, measured on both training and validation data.
EarlyStopping monitors validation loss; training stops after 3 epochs without improvement and restores the best weights.
ModelCheckpoint saves the model with the best validation accuracy as best_mask_model.h5.
The final trained model is also saved as mask_classifier_model.h5.

# Prediction on a New Image

WHAT PREDICT.PY DOES

## 1. CHECK FILES

Confirm model and image paths exist.

## 2. LOAD MODEL

Load best_mask_model.h5.

## 3. READ IMAGE

Open image with OpenCV; handle unreadable files.

## 4. PRE-PROCESS

BGR → RGB, resize to 224×224, scale to 0–1.

## 5. PREDICT

Add batch dimension and call model.predict().

## 6. DISPLAY

Choose class using 0.5 threshold and print confidence.

# Prediction Logic & Confidence

HOW THE OUTPUT IS SELECTED

```
prediction = model.predict(img)[0][0]

if prediction < 0.5:
    label = "with_mask"
    confidence = 1 - prediction
else:
    label = "without_mask"
    confidence = prediction
```

## THRESHOLD = 0.5

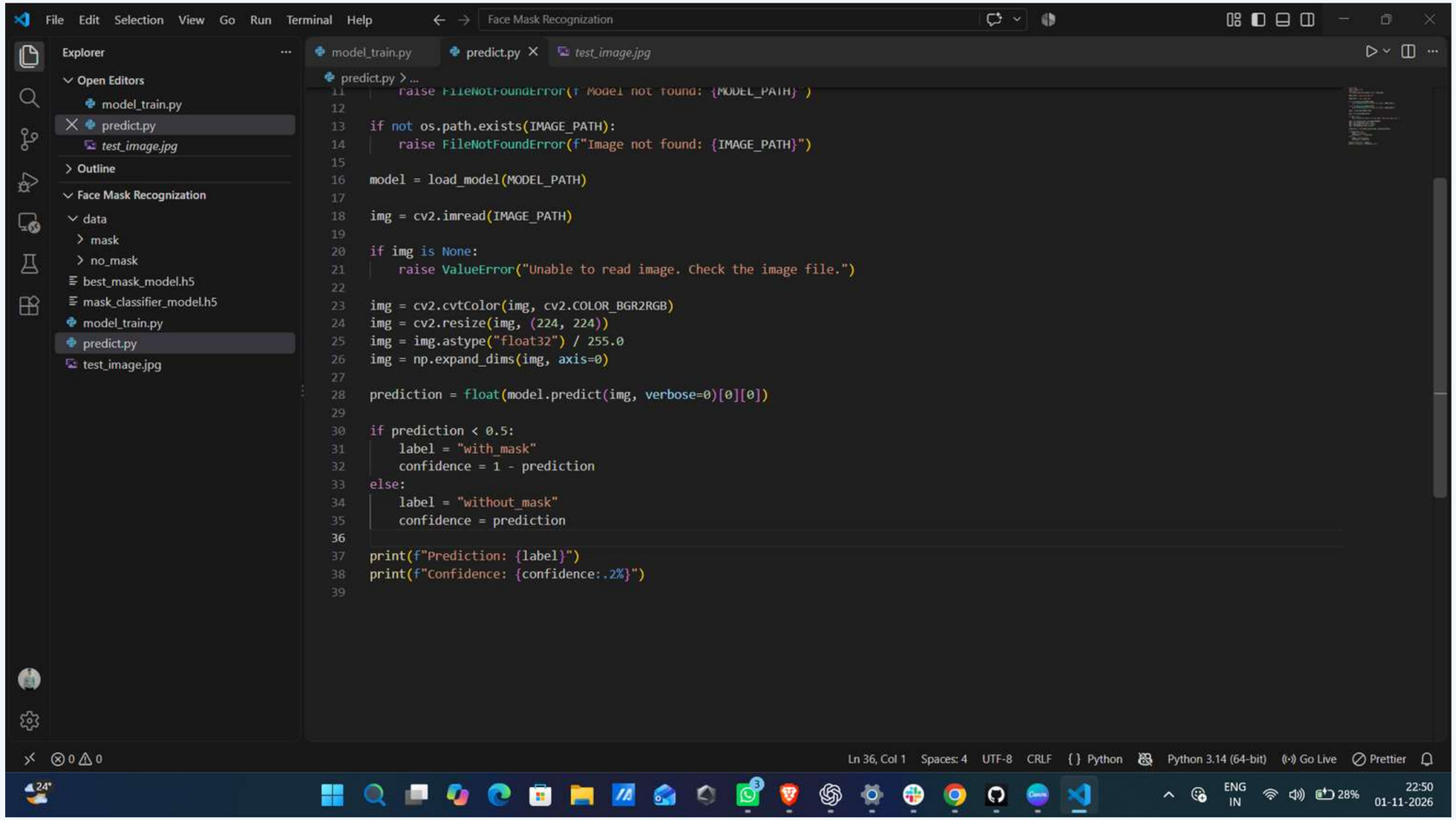
Below 0.5 → With mask
0.5 or above → Without mask

## CONFIDENCE SCORE

The displayed confidence is the model score for the selected class. It is not a guarantee that the prediction is correct.

# Implementation & Output

PROJECT IN VISUAL STUDIO CODE




## PROJECT FILES

model_train.py
predict.py
data/
best_mask_model.h5

## OUTPUT

Prediction label and confidence are printed in the terminal.

# Limitations & Future Improvements

WHAT CAN BE IMPROVED NEXT?

## CURRENT LIMITATIONS

- Predicts one image at a time.
- Performance depends on dataset quality, size and diversity.
- Image classification does not locate multiple faces separately.
- Confidence score may not equal real-world accuracy.

## FUTURE SCOPE

- Build a web interface to upload an image from a browser.
- Add face detection for images containing multiple people.
- Test on a larger and more diverse dataset.
- Fine-tune selected MobileNetV2 layers and evaluate performance.
- Add webcam/live-video support with appropriate privacy safeguards.

# CONCLUSION

The project demonstrates how transfer learning with MobileNetV2 can be used to classify face-mask images into two categories.

It combines dataset preparation, model training, evaluation and prediction on a new image.

## THANK YOU

Questions?